

Tell the story first. Let the map be inspected second.

A guided mental model of Azure SPI, backed by an inspectable architecture map. This is the record of what the site is becoming, why, and how each remaining piece changes. Every lesson teaches with the drawer closed, offers optional evidence and an optional **Try it** activity on the learner's own account, and compares against CIMPL, the community implementation, where the comparison teaches. The current round makes the site easier to read straight through: one index on Start, the video in a modal, the selected idea beside its map, and fewer invitations to leave a lesson. The rest is planned here and tracked in beads.

State 14 September 2026, main at 62be86e: lessons 01 to 03 live with recipes; the reading-flow round shipped; Start, Audio deep dives, and Visual field guides reworked with the owner; the polish review closed; the Learning design review shipped: idea selectors inside the map workspace, the claim counter gone, the inspector flattened, Go deeper as a quiet list, and lesson 07 presented as a field check; the Explore map toggle is gone, closing the drawer keeps the map selection, and the field-guide section chips no longer rotate; lessons 01 to 03 end in one Go deeper shelf of closed, illustrated rows under an illustrated Try it banner, and every chapter exits through a footer pager instead of a dark next bar; lessons 04 to 07 carry the same footer as 01 to 03; the deep-dive player no longer links out to the notebook or a download; Visual field guides sections are badged rows with previews that open onto a card grid; lesson 06 is driven by its run steps, and the tracker holds four owner items and one review pass. Where an earlier section describes a decision this line supersedes, the section says so and the changelog holds the history.

Plan of record beads in the repo (`bd list`); this page explains the intent behind each bead

SHIPPED

The reading-flow round, on top of lessons 01 to 03 with evidence, CIMPL, and Try it

Live on the published site. Start: no rail, one grouped index, the stack-and-engineering promise, the video in a modal, a Resources disclosure instead of Explore and Listen in the masthead (a9a5ece; historical: revision 12 replaced the disclosure with two quiet icon links, Audio deep dives and Visual field guides). Lesson 01: idempotent selection, the selected claim beside the map, no question or goal row, the example collapsed after the exit on every lesson, the shorter opening (PR #14, 9575c34; phone diagram top 1,128px from 1,547). Lesson 02: the stage bar as the one control, claims as statements, a single Continue that lands the next explanation under the sticky bar with focus, evidence links that stay in the lesson, a one-line Try it summary (e4a8292). The editorial sweep: kickers and vague references gone with a test that pins the phrase list, chips reduced to label and minutes, posters in the lightbox with their notes, the profiles guide opening in place (a442848). Earlier the same day: the seven round-five fixes, CIMPL, the recipes, and the environment name as a placeholder.

NEXT

.....

One review pass, then three owner tasks

The site is at its done state for the build: every lesson carries the frame, lesson O6 is step-driven, and the three pages match. What remains is a reading-flow check on lessons O3 to O6 in their final form (fn-kwp), then the owner's three: the personal-account fork walkthrough (fn-dt3), whose outcome is the O4 and O5 bands and the promotion of O2's preview; the pilot (fn-29r); and the recordings from the approved scripts, including the brief (fn-y2o).

THEN

Only on evidence

Claim selectors for lessons O4 and O5, a CIMPL field guide, or any navigation change reopen only if the pilot or the reading-flow pass shows a learner stopping. Explore is closed (fn-3v8, not doing).

The experience

Learn tells the story.

Seven sequential lessons. Each opens with its claims, shows them on the map, and closes by handing the same claims back as things the learner can now say.

The map illustrates the story.

One map per lesson, at full width. The current claim decides which components are at full weight. Everything else recedes but stays clickable.

Explore unlocks the map.

The same map with every component open, as a switch on any lesson today and as its own page next. No second diagram to maintain.

The inspector supplies evidence.

A drawer, closed until asked. What it is, why it matters here, how to verify it, where to go deeper. One template for every object, every easy mistake, and every entry in lesson O7.

Try it applies the lesson.

An illustrated banner after the bridge (revision 19: badge, title, one-sentence outcome, Optional qualifier, on a pale blue ground): what to run or open to see the lesson's idea in a real repository or environment, with what to look for and what else might correctly happen. The site describes; the workstation runs. Nothing is tracked or scored.

Lesson 01, live →

Lesson 02, live →

Lesson 03, live →

The prototype it was built from

PR #4

The rule that keeps it from drifting back

Nothing required to understand a lesson lives only in the inspector. The page answers "what am I supposed to understand?"; the drawer answers "how do I prove, check, or debug that?" If a fact in the drawer is needed to satisfy "What you can now say", it moves to the lesson surface. If a fact on the surface only answers how to verify something, it moves to the drawer. This is a review rule for every content change, recorded in AGENTS.md. The litmus questions in section 7 are how we check it, and the pilot is where we check it first.

- 1 · Rules every lesson follows
- 2 · What shipped in lesson 01
- 3 · Layers and the evidence template
- 4 · Lessons 02 to 07, lesson by lesson
- 5 · Try it: the hands-on track
- 6 · Navigation, Explore, and the rest of the site
- 7 · Litmus questions and the pilot
- 8 · The plan in beads, and how work runs
- 9 · Decisions
- 10 · Changelog

Section 1

Rules every lesson follows

Recorded as beads decision fn-nh5 and in AGENTS.md. A change that breaks one of these is a change to this document first.

- 1 **The lesson grammar carries responsibilities, not rows.** Position line, headline (the sentence to remember), one short lead paragraph that carries the concept, then one map workspace: its title row, two to four idea selectors across the top, the selected idea's explanation directly under them, and the map beneath; then optional evidence in the drawer, one easy mistake, "What you can now say" under one heading, the bridge to the next lesson, then **Try it**, then one Go deeper shelf: full-width disclosure rows, all closed, in the order Follow the example, Hear it explained, Field guides, Sources, each present only when the lesson has that content, drawn with a small field-notes badge, a bold title, and a one-line preview counted from the content (revision 18, 19, and 21, when the shelf, caption, light outcomes, and pager reached lessons 04 to 07 ahead of their claim grammar; lessons 04 to 06 keep the older below-figure layout until they roll into the grammar). The question, goal, and example stay in the content model for review; they are not each displayed as a row of interface text. A lifecycle lesson uses its stages as the workspace control, with its claims as concise statements after the map. Revised in revision 11 from "fixed" after the reading-flow review, and in revision 16 to put the selectors inside the workspace (the Learning design review found the control and the map it changes could be a screen apart on a phone).
- 2 **The claim is the primary interaction, and selecting it is idempotent.** Selecting a claim changes the map and the explanation directly above it; selecting the selected claim leaves it selected and opens nothing. The selectors carry no counter: three ideas are not three steps. In a lifecycle lesson the stage is the primary interaction and the map, the

three steps. In a lifecycle lesson the stage is the primary interaction and the map, the stage explanation, and its evidence entry change together. Everything else on the page is evidence for a claim, except the optional practice in Try it, which applies the claims rather than proving them.

- 3 **A claim has three parts.** A short headline (the control, under about twelve words), the full sentence (what "What you can now say" hands back), and the reason underneath.
- 4 **"How we know" opens the drawer on the best evidence item** for the selected claim. That is the intended way into the drawer during a lesson.
- 5 **The running example is optional depth.** It is the first row of the Go deeper shelf, "Follow the example", collapsed to one line until asked. Expanding it and selecting a hop enters a trace: the path up to that hop at full weight, the drawer on the component it reaches. It no longer sits between the claims and the map.
- 6 **Focus is the only map behavior.** Only the evidence component shows "Inspect ↗". Other components recede but stay clickable, with the affordance on hover or keyboard focus, and on a device without hover they recede less so they stay readable. The Explore map toggle was removed in revision 17: it changed only weight and affordances, nothing was reachable only through it, and the owner found it implied a hidden layer.
- 7 **Focused components at full weight, the rest recede.** Boundaries (the stack, AKS, the service) never recede; a faded container reads as "gone", which is what the teardown state means.
- 8 **The drawer is closed by default and opens by intent:** a component click, a hop, an evidence link, an easy-mistake map jump, or a route carrying `?detail=`. It closes with `x` or Escape and returns focus to what opened it.
- 9 **The drawer has a lean template:** the component name; one small line of context, with the owner after a separator only when one is authored; two plain paragraphs (what it is, why it matters here) with no labels between them; More; Verify it; Sources. The labels Inspecting, Context, Owned by, What it is, and Why it matters here were removed in revision 16 because together they read as a form. The context line is metadata, never an owner, and "Why it matters here" is a short answer written for the current lesson claim, with chosen depth after it (fn-p2j, shipped). The selected claim reads beside the map with the drawer closed; the drawer is evidence, not the lesson. Closing the drawer keeps the selected component and any map state it drives, so lesson O6's lock and pod stay where the reader put them (revision 17, from the lesson O6 review); selecting a claim, a stage, or another lesson clears it.

- 10 **The lesson works with the drawer closed.** Nothing required for "What you can now say" lives only in the drawer. A detail that answers "how do I prove, check, or debug this" belongs in the drawer; one that answers "what should I understand" belongs on the page.
- 11 **The lesson closes, then the extras, then the exit.** Easy mistake (one last test of the mental model, explained locally, with no link to the whole collection and no evidence link that changes chapter), What you can now say. Then the Try it banner. Then the Go deeper shelf. Then the pager at the page end: the position line, a quiet Previous, and a Next lesson card with the next lesson's title and question (revision 20; the dark full-width Next bar between the outcomes and Try it was retired because it read as a panel rather than as part of the page, and a docs-style pager is the exit readers already know). The three-column band with the poster card, the separate example row, and "Open here" were replaced in revision 18 because they asked the reader to decode four treatments after the exit and made the page a quarter taller; the poster is now a title, a sentence, and View poster inside the Field guides row, and the scope sentence moved up to a caption under the figure.
- 12 **The headline is the sentence to remember.** The lesson name lives in the rail and the position line ("Lesson 3 of 7 · One service").
- 13 **Historical: the local toggle read "Lesson focus | Explore map"** until revision 17, when it was removed. The masthead's Explore wording is no longer at risk of confusion with a switch.
- 14 **"Listen to this lesson" waits for a recording that is the lesson.** Until a narration is recorded from the claims, the cue row is labelled "Hear it explained" and is a shelf row whose play chips sit inside the opened row, so every row is one action (revision 19; revision 18 had the chips beside the row, which the reviewer read as two actions on one line).
- 15 **Nothing removes or renames a route.** Deep links from markers, field checks, and posters keep resolving; a link with `?detail=` opens the drawer.
- 16 **Interaction rule.** Selecting an idea or stage produces a visible local result; opening media or evidence preserves the reading position (the video opens in a modal, the drawer overlays, a poster opens in the lightbox with its notes); a control that navigates names its destination ("Show it on the map ↑", "Continue to Use the stack →") and brings what it changed into view. Deliberate navigation may move the page and says so; nothing else does. This replaces the earlier "nothing moves the page".

- 17 **Try it is optional and states its prerequisites.** Each activity can be started once its prerequisites are met and links to shared setup rather than repeating it. Its collapsed line says what the learner will do; expanded, it names the intended result, the access it needs (browser only; workstation setup; public GitHub repository; Azure resources billed separately), active effort separately from automated waiting and clean-up, each step with what to look for, the common alternate result and what to do then, and the clean-up with what remains afterwards. The site never runs a command and never shows live state; a browser-only route ships once it is verified in the browser; an Azure route ships only after someone has walked it and recorded the tested versions, and until then appears only as a clearly marked preview (lesson O2 today). The learner's own environment name replaces `<name>`; `opendes` stays as the example partition; no environment of ours appears. Section 5 has the track.

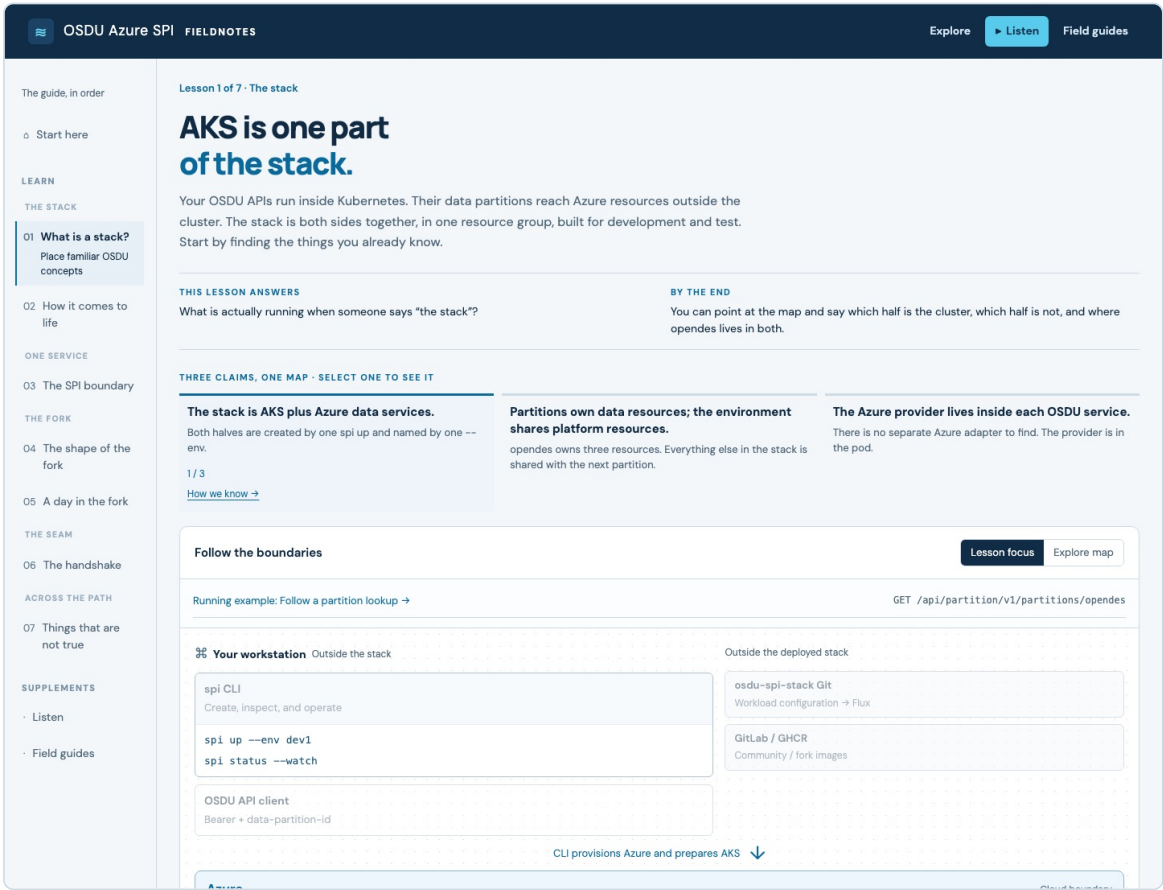
Section 2

What shipped in lesson 01, and the fifth review

Round five (13 September, `docs/lessons-01-03-implementation-review-2026-09-13.md`, `bead fn-2bc`) reviewed the three published lessons at 1440 and 390 wide, every claim, every evidence link, all fifteen hops, and both lesson O3 trace conditions. Verdict: keep the structure and the visual direction. Seven follow-ups, all accepted:

- **fn-kbc, P1.** Lesson O3's resource block puts Redis and common Table Storage under one "Workload Identity" label. Redis runs inside AKS with middleware credentials; Table Storage is outside AKS on Azure identity. Draw them as two dependencies crossing two boundaries.
- **fn-loo.** Lesson O2 calls CLI success "the first of five milestones". The CLI verifies the Git source before it returns and Flux overlaps its last stages. Describe distinct readiness signals, not a sequence.
- **fn-mjx, fn-5pw, fn-wes.** Three reproduced drawer defects: at 1440 × 800 the drawer opens with 24 px visible; first opening leaves focus behind the drawer; reloading lesson O3's provider evidence URL turns a claim into an expanded trace.
- **fn-p2j.** Evidence that answers a different question: the "one image" claim leads to registry naming; the proof is the provider POM's dependency on partition-core and the Dockerfile. "Owned by" often holds a place, not an owner; that is a content field in the shared evidence component.
- **fn-e7u.** Copy: author-facing rules in learner text ("with the drawer closed"), counts as formulas, and fork branch names introduced before lesson O4.

Also noted, for the map: the components a claim changes sit a full viewport below the claims on desktop and further on a phone. A compact selected-claim reminder beside the map, or a shorter claim area, is the candidate fix; ordinary selection still must not move the page.



The published lesson 01 at 1440 px after the round-four tunes: short claim headlines with the reason under each, and the rail showing a subtitle only on the current lesson.

- 1 **Position line** "Lesson 1 of 7 · The stack" replaces the kicker; the premise line is gone.
- 2 **Two-column orientation:** the question and a new one-line goal. "You are here" and "See the six places" left the lesson; the six places stay on the start page.
- 3 **Three claims** from the existing outcomes. Each is a short headline over its reason; the full sentence returns in "What you can now say". Selected state: subtle background, a 1 / 3 marker, and the "How we know →" link. Each claim carries its headline, text, why, focus, scopes, evidence, and crossing label in chapters.js , so a second lesson with claims needs no code.
- 4 **The map at full width**, focused by claim. Receded components at 45 percent, still clickable.
- 5 **Running example collapsed** to "Running example: Follow a partition lookup →" with the GET line; hops are a line of type, and each one traces.
- 6 **The drawer** on every map lesson, template as specified, phone sheet retained. Escape and × return focus. A deep link opens it without stealing focus from the headline.
- 7 **Easy mistake as a rule, carry forward, Next**, then Try it and the Go deeper shelf (revision 18; earlier a three-column band with cues, guide previews, sources, and the scope note).
- 8 **Lessons 02 to 07** keep their page shape. Their nodes show the Inspect label on hover and focus only, and no component is selected on load.

- 9
- Round-four tunes (4b7ed98): claim headlines shortened to one line each, rail subtitles shown only for the current lesson, and the tests now hold every structured lesson to two to four claims with a headline shorter than its sentence.

VERIFIED BEFORE MERGE

- `npm run check` : formatting, 13 content tests including the new claims check (every focus, scope, and evidence ID resolves in the rendered scene on both overview paths), production build.
- Headless screenshots at 1440 and 390 px of lesson 01 (both paths, with and without a detail), lesson 02 at Assemble OSDU, and lesson 06, compared against the prototype.
- A second-agent review of the diff found ten defects; nine are fixed (stale trace after Back, drawer clipped on tall maps, `aria-pressed` on the diagram element, focus stolen from the headline on deep links, unannounced drawer, hardcoded lesson text, modifier-click state leak, claim not inferred from a bookmarked evidence route, brittle test). One is accepted by design: closing the drawer rewrites the URL without its detail, so one Back press after closing does nothing visible.

Section 3

Layers and the evidence template

Story, map, inspector, source. Every piece of content sorts into one, and the data files keep them apart. The page decides which layer is visible by default.

LAYER	ANSWERS	WHERE IT LIVES	DEFAULT VISIBILITY
Story	What am I supposed to understand?	chapter headline , intro , question , goal , outcomes (as claims); moment title and copy ; example note	Always on the page
Map	Where is that, and what is it next to?	the five renderers; claim focus and scopes ; example hops ;fork-day active lanes	Always, focused by the claim
Inspector	Tell me more about that thing.	componentDetails ; the myths; the marker source-check notes	On request
Source	Where is that substantiated?	sources.js , 60 entries	Last line of the drawer; end of the page
Practice	How do I see this in my own	chapter tryIt (planned, fn-u9u): result, prerequisites, success effort and waiting steps with	Collapsed line after

in my own repository or environment?

prerequisites, access, effort and waiting, steps with observations, alternate results, clean-up, tested versions

time after
Next;
expands in place

The drawer template, as first shipped (revision 9)

Historical: the reading-flow round since put the selected claim beside the map and dropped the question and goal rows; the shipped block at the top and section 2 describe the current drawer.

LINE	FROM
FLUX	Inspecting
the node's title	Flux
Owned by	label
	flux-system / osdu-flux
What it is	the node's subtitle
	Reconcile workloads
Why it matters here	title + first sentence of body ; the rest behind More
	Git fetching and reconciliation are separate switches. The Git source is suspended after deployment by default.
Verify it	artifact.label + artifact.code
	Read rollout conditions · kubectl get kustomizations -n osdu-flux
Go deeper	source
	Flux dependency ordering ↗

One evidence component, four places

The myths already have the drawer's shape: claim , reality , check , source , route . The details have it: title , body , artifact , source . The marker notes have the middle two. The roll-out epic renders all of them from one component, so "here is the evidence" looks the same whether the learner met it on a map, under a map, on lesson O7, or beside a recording. Two content follow-ups ride along: an editing pass that trims each detail body to the sentence that matters (the fold is the safety net until then), and an optional here override per chapter for the few details shared between scenes (client in O1 and O3,

image in O5 and O6).

Section 4

Lessons 02 to 07, lesson by lesson

The reference implementation is CIMPL. "You know OSDU" names the audience but not what they compare the Azure implementation against. The start page introduces CIMPL, the open-source community implementation, in one paragraph, and the lessons use it where a difference explains the Azure side: the provider code inside a service, the resources that code reaches, who owns a change, and what a deployment test adds to a community test. Three subjects stay separate: the service implementation (partition-core-plus with PostgreSQL versus provider/partition-azure with Azure clients), the running environment (cimpl-stack versus osdu-spi-stack), and the engineering workflow (the community repository versus the synchronized fork). Two facts shape the wording: CIMPL can run on AKS, so hosting does not select the implementation; and Elasticsearch, Redis, and Airflow's PostgreSQL stay inside the Azure stack, so there is no universal product-substitution arrow. The paired partition lookup (VmCache then PostgreSQL, against Redis then common Table Storage) is the one comparison that earns a diagram, as an optional "Compare the partition lookup" disclosure on lesson O3. Everywhere else it is a sentence of 25 to 45 words replacing generic prose, with source evidence in the drawer. The review is `docs/cimpl-comparison-review-2026-09-13.md`; the prototype is `fn-jy3`.

Each lesson is written backward: what should the learner be able to say, which claims establish that, which map state proves each claim. For O4 to O7 the claims below are the lessons' outcomes as they stand today, to be corrected during the roll-out where a review has already flagged the wording; the focus and evidence IDs are the components that exist on each map today. For O2 and O3 the rows are historical: the shipped and corrected wording is in the repository. A candidate headline is given where the current one is a description rather than the sentence to remember; those are content decisions for the roll-out, not settled here. Each claim also needs its short headline, written when the lesson is done; the sentences below are the carry-forward lines. Lessons O2 and O3 went first and the grammar held; the rest follow the same brief, now with a Try it band each (section 5).

02 How it comes to life

Shipped Headline: **One command creates the environment. Flux finishes it.** PR #5.

The six moments stay the sequence control. Each claim carries an entry moment, the moments it covers, and the moment where its evidence exists (claim 1 enters at Create Azure and proves itself at Assemble OSDU, where Flux is). Selecting a claim moves the lesson to its moment; the old per-moment auto-open is a "Look closer ↗" link under the moment story. Readiness became a real component at Use the stack so its evidence link resolves. Claim headlines: "CLI creates; Flux assembles; controllers keep workloads

resolves. Claim headlines: "CLI creates, Flux assembles, controllers keep workloads healthy.", "CLI success is not API readiness.", "Teardown removes data but preserves identities and names."

- 1 The CLI and Bicep create Azure and seed the cluster; Flux assembles the workloads; controllers keep them healthy. Different owners, different clocks.

moment Create Azure → Assemble OSDU · focus workstation , aks , bootstrap , flux · evidence flux (the Kustomization conditions)

- 2 ~~A successful spi up exit is the first of five milestones, not readiness.~~ Corrected by fn-loo to: A successful spi up does not establish API readiness; follow workload health and initialization with spi status --watch, then verify the API path with an authenticated request.

moment Use the stack · focus readiness , initialization , caller · evidence readiness

- 3 spi down removes compute and data but keeps identities and the resource group, so a rebuild reuses the same names.

moment Remove the stack · focus retained · evidence retained (az resource list on the footprint)

03 The SPI boundary

Shipped Headline: **The provider lives inside the service.** PR #6; boundary corrected in PR #7 (two lanes, six hops); the CIMPL comparison disclosure added in PR #12.

The cache-down trace is a Normal / Cache down switch inside the running example: one five-hop trace, two presentations, hop four "Cache read throws; treated as a miss" and hop five "Table Storage answers anyway", with the same text on the map. The SPI map gained the image boundary, the upstream and engineering-system nodes, and a source-ownership panel beside the runtime column. The Azure implementation's Go deeper list is the fc2dfbf fix, then ADR-038, then the provider class, and the drawer names the real regression test. The cache-throw fact is in claim 1's reason on the page. Claim headlines: "Common code calls Azure through a provider interface.", "The interface and implementation ship in one image.", "The fork keeps Azure source outside the generated tree."

- 1 Common service code calls a provider interface; the Azure implementation behind it does the Azure work with Workload Identity. For the partition service that is a cache, then a Table Storage read, and the read still happens when the cache throws.

focus core , contract , azureimpl , azureclients · evidence azureimpl

- 2 The interface and its implementation ship in one image. There is no network hop between them.

focus contract , azureimpl , image · evidence image

- 3 Upstream may delete its Azure implementations; the fork owns provider/<svc>-azure and keeps it outside the generated upstream tree.

focus upstream , azureimpl , engineering · evidence upstream

04 The shape of the fork

Headline now: **Upstream plans to remove the Azure code. The fork is where it lives.** Candidate: **The fork owns the Azure code.**

The ownership table is the map and its what-if switch is already a policy. A claim emphasises rows or columns; claim 3 turns the switch on. The fork-owned rows come first in the focus so the minimal core-versus-provider contrast is what the learner sees before the engineering lanes.

- 1

The fork owns provider/partition-azure, its Azure tests, its descriptor, and the engineering files the template delivers. Everything else is upstream's, regenerated daily.

focus rows provider-azure , engineering-files , the descriptor row · evidence provider-azure
- 2

The fork has three branches with three jobs: fork_upstream is generated input, fork_integration is the workspace, main is the protected result. Upstream is outside all three.

focus columns upstream , fork-upstream , fork-integration , main-branch · evidence main-branch
- 3

fork_upstream is generated with the Azure paths absent by construction, so an upstream deletion has nothing to delete on the fork side.

turns the what-if switch on · focus fork-upstream , provider-azure · evidence filter

05 A day in the fork

Headline now: **From an upstream update to a candidate image.** Candidate: **Nothing reaches main without a person.**

Already has focus: each moment names its active lanes. The change is the page shape, the drawer, and one bridging sentence at the top saying how the provider fix reached main (today that story is told in lesson 04's hops). The five moments stay the primary control; a claim selects its moment.

- 1

A sync is a generated tree plus one PR and one tracking issue; upstream is never merged in as text, and the generated commit lands through a reviewed sync PR. The cascade merges main first, then fork_upstream, into the workspace.

moments Generate, Integrate · evidence sync-pr , then cascade-run
- 2

The labels on the tracking issue are the state: cascade-active, cascade-blocked, cascade-failed, validated. A blocked cascade is run again after the fix on fork_integration; removing human-required is how a failed one retries.

moment Propose · evidence labels
- 3

The integration PR and the version PR are different PRs. Validation runs on PRs and on pushes to main and fork_integration, pushes a digest for every eligible commit, and the stack is borrowed for it before any release exists.

moments Build and prove, Release · evidence candidate

06 The handshake

Headline now: **Write the lock. Flux rolls out the image.** Keep it.

Sparse by rule; the hops are the prose. The lock-and-pod state panel stays on the map and repeats at the top of the drawer, as the phone sheet did before. The six hops are the trace; the three claims are the three things the state panel proves.

As shipped in revision 23: the six steps sit in the selector slot as buttons, three per row on a phone; a step moves the state panel and focuses its component without opening the drawer; the three claims render as statements after the workspace; the Follow the example row is not rendered because the steps are the control. A step is `#handshake?hop=n`, its evidence `#handshake?detail=<id>&hop=n`, and every published `?detail=` link for a step's component still lands on that step with the drawer open.

- 1

Writing the lock starts the deploy: Flux reconciles osdu-image-lock and the pod restarts on the pinned digest, and the run checks the pod before trusting any test result. Between runs the lock holds the canonical image.
`focus delivery , running , verify · evidence delivery`
- 2

The run only restores what it still owns, so a green restore is a claim about this run, not about the environment.
`focus restore , lock-taken · evidence restore`
- 3

The stack publishes facts and the fork declares needs in its descriptor. Five repository settings, plus trust onboarding on the stack side, are all that connect them.
`focus facts , descriptor , trust · evidence descriptor`

07 Things that are not true

Headline now: **Field check: the assumptions that cause trouble.** Revised in revision 16.

Stays a lesson in the index and rail, and reads as a field check after the round trip closes on 06. Its 22 entries render as rows grouped by theme, each naming the lesson it was built in: the quoted assumption on the left, a one-sentence correction on the right, and the full explanation, the check command, and the two links inside a closed "Check and sources" disclosure. Every entry keeps its anchor, so easy-mistake callouts under the maps still resolve. No claims strip; the page is the evidence.

Stays as built across all seven.

The lesson sequence and its questions. The running example and its hops. The boundaries map and the shared renderer. Easy mistakes under the map. Owner colours, orange for the fork. Persistent audio and cues that play in place. The start page's three meanings and six places. No quizzes, no scores, no completion tracking; "Lesson 3 of 7" is a position, not a progress bar.

... and a preliminary note page...

Section 5

Try it: the hands-on track

The lessons explain and show; nothing yet lets the learner do. The fix is one optional band per lesson, after Next and before the optional depth, that says what to run or open and what to look for. Everything below was checked against the checkouts named in README's "Reviewed against" and against the source-checked review of the first draft (docs/try-it-plan-review-2026-09-13.md , fn-wqb). The GitHub activities can be tried without Azure resources; the Azure ones say what they create and what they cost.

Each activity states its prerequisites and can be started once they are met. The conceptual lesson order stays. A reader without Azure gets a quiet practical entry: browser inspection, workstation preparation, then the public GitHub route. Activities link to shared setup rather than repeating it. Each one applies its lesson's idea; where the natural activity is preparation for the next lesson, it says so.

LESSON		COLLAPSED LINE	INTENDED RESULT	STEPS AND WHAT TO LOOK FOR	ALTERNATE RESULT
01	Live	Try it: prepare your workstation	The five tools are installed and you can tell tool readiness from Azure access. This is preparation for lesson 02.	Install uv , then the CLI from the stack's GitHub release with uv tool install ; run spi check and read spi up --help for --profile , --partition , --location . Look for the five tools checked (az, bicep, kubectl, kubelogin, flux) and notice that nothing signed you in.	A missing is named the check install it rerun.
02	Live Azure route not walked	Try it: bring up an environment	An environment you own, its readiness signals read separately, and one authenticated partition request.	Two routes. Without Azure: read the documented example output of a run and of spi status --watch , focused on stable fields. With Azure: az login and confirm the subscription; spi up -env <name> --dry-run (this needs Azure access and creates or	Provision can fail (regional Automate capacity resource group m already and need same cl up. Dow waits up

updates the resource group and its naming tag, then previews the templates); the real run on the core profile with a tested `--tag ; spi status --watch ; spi info --show-apis ; spi token` and one authenticated lookup of a partition. Then choose: keep the environment for lesson 06, or `spi down --env <name> now` (identity and naming retained) and `--purge` when done for good.

03	Live	Try it: trace the partition lookup	You can follow <code>getPartition</code> from shared code into the Azure provider, through <code>safeGet</code> on the cache to the table fallback, in the real source.	Open the provider's <code>PartitionServiceImpl</code> at the reviewed revision and follow the call; then the provider POM's dependency on <code>partition-core</code> and the Dockerfile's single image. Optionally run the fallback regression test once its build prerequisites are documented. With a stack up: list the partition deployment and identify the image it runs, then say whether it is a community image or one a fork built.	A fresh s runs commun images; fork-bui candida appears in lessor
04		Try it: create your own fork	A syncing fork on your personal account whose ownership and branch structure you can inspect.	Register a personal GitHub App with repository permissions and install it on the new repository; use the <code>Azure/osdu-spi</code> template; set the two secrets; answer the init issue with the upstream URL. Look for the init issue within a minute, Initialize Complete in	Actions need en by hand inheritec CODEO\ with Azu handles validatic must be replacec

five to ten, then
fork_upstream ,
fork_integration ,
and main . Check the
service package's
visibility settings now,
before any Azure step.
The mirror alternative
(fork Azure/osdu-spi-
partition, dispatch
Adopt Fork) is a
different route with its
own gotchas and is not
equivalent for lesson O6.

O5	Try it: inspect an upstream sync	You can read a sync result, including the case where nothing needs updating, and see one controlled provider change validated.	Dispatch Sync Upstream by hand. Already current: the run exits cleanly with no PR; that is correct. New shared changes: inspect the sync PR and its tracking issue, merge, and watch the cascade open the release PR into main. Then make the named change under provider/partition- azure from the walkthrough (a specific file, branch, and PR target, with the expected build result) and watch validation build it and push a digest.	If no PR appears check w the fork already current. default r needs a approval review a authors cannot approve own PR: observir PR's vali is a com stopping point; m your own change i a second reviewer
O6	Try it: deploy a candidate to your stack	Your fork's candidate digest reaches a ready pod in your environment, meaningful tests run, and the lock is restored.	Planned, not written, until the route is demonstrated. It needs spi onboard partition --repo <you>/<fork> (plan, then --write) on a core-profile environment, a workflow revision with the deploy-test lane, a tested	The refe partition at the reviewer revision no desc and not newer la Clean-u onboard remove then wri

.spi/service.yaml for does no
partition, consistent remove
service identifier and reposito
package name, a public own val
package, and an eligible
change that builds an
image. Look for the gate
entering the deploy-
test lane, the digest on a
ready pod, at least one
non-skipped test with
no failures, and the
restore result. A green
overall check alone can
mean deployment was
skipped.

07	none	The page is the reference; no band.
----	------	---

What the repositories require of a personal account

- **No organization.** The docs say "your organization's GitHub App" and "admin of the GitHub organization" (adoption.md, fork-lifecycle.md, adopt-fork.yml), but the App uses repository permissions only, the retention workflow handles user-owned packages, and the roster design names personal forks. The band writes the personal-account path; the org path becomes a note.
- **Two secrets, no PAT, no GitLab token.** RELEASE_APP_ID and RELEASE_APP_PRIVATE_KEY . The "Personal Access Token with workflow scope" message in init-complete.yml is a stale error string.
- **Review rules apply to the owner too.** The default ruleset requires an approving review with an empty bypass list, and GitHub does not let an author approve their own PR. App-authored sync and cascade PRs can be approved by the learner; the learner's own change cannot be merged alone.
- **A pushed image is not necessarily pullable.** The build's visibility helper reports a private package and neither changes it nor fails the build; some descriptions still say it flips visibility. AKS then fails with ErrImagePull. Package visibility is a step in lesson 04, before any Azure work.
- **Connecting a fork to a stack is spi onboard , not Flux.** Flux always points at the stack repository. A fork's image reaches the cluster as a digest pin written by the fork's validation run after onboarding, and a successful test restores the canonical image afterwards.
- **The cheap profiles cannot host lesson 06** Onboard refuses minimal and bare

The cheap promises cannot meet request for ensure a release minimal and safe.

- **No conflict is resolved by an AI.** ADR-005 rejects automatic resolution; Copilot appears only in optional review and a setup workflow.

Gotchas the bands name

- Forking disables inherited workflows; scheduled ones sit disabled until Adopt Fork re-enables them. Enable Actions on a new repository by hand.
- A fork named osdu-spi-partition needs `SERVICE_NAME=partition` or package resolution mismatches.
- `spi check` checks tool presence only. Sign in with `az login` and confirm the subscription before `spi up`, including before `--dry-run`.
- Regions differ in AKS Automatic capacity; westus3 is the default and eastus2 and centralus are documented as constrained. The CLI has no quota preflight.
- No cost figure is published. A band says "Azure resources billed separately" and puts the keep-or-remove choice in the same step as the run.
- Contribution PRs from a mirror fork back to Azure/osdu-spi-partition need a maintainer there to approve workflow runs. The bands stop at the learner's own fork.

Out of bounds

`spi service refresh`, `--canonical-source`, and the stale-sweep workflow step are documented ahead of the code. No band uses them. Lesson 06's band stays planned until someone has demonstrated the exact descriptor and workflow route.

How the band is built

- Content: a `tryIt` field per chapter in `chapters.js`: the collapsed line, intended result, prerequisites with links to shared setup, access label, effort and waiting and clean-up as separate figures, steps (each a command or click plus what to look for), alternate results with the next step, clean-up with what remains, and the tested versions (CLI release, stack ref, template commit, shell, date). Placeholders are marked and the shell is named. Sample output is marked illustrative and shows stable fields only. Tests check the structure and that every link resolves.
- Rendering: one component after Next, collapsed to its line, expanding in place. It does not move the page, opens no drawer, stores nothing, and shows the access and resource effects before the first command.
- Order: the renderer and tests shipped in PR #13 with no production band; lesson 01's recipe followed on main after `spi check` was walked on the author's machine (`spi` 0.16.0 against `dc2c956`; the install command is quoted, not re-run). Next the recipes for 02 and 03 (`fn-z1m`) with browser-only and Azure-backed variants explicit, then 04 and 05 written from the walkthrough, then 06 when its route is demonstrated.

- Proof (fn-dt3, yours or a volunteer's): the personal-account walkthrough covers App and repository setup, an unchanged sync, a changed sync when available, an author-created PR against the review rules, package visibility, and clean-up, with timings. The lesson O6 walkthrough additionally records the descriptor, workflow revision, candidate digest, non-skipped suite results, and the restore result. Content tests and browser checks prove the page, not the instructions; the walkthrough does that.

What does not change. No required exercise, no score, no completion state. The lesson still works with the band collapsed and the drawer closed. The site executes nothing, reports no live environment, and shows no token, key, or account state. The running example is `opendes` in the environment the learner names; every command carries `<name>` where that name goes.

Section 6

Navigation, Explore, and the rest of the site

Navigation frame (reopened in revision 11)

Reopened by the reading-flow review. Start had 43 links across overlapping representations of the curriculum (rail, doors, cards, round-trip stops, SPI-meaning links, the six-place ladder), a masthead Explore that opened lesson O1 under another name, and Listen and Field guides offered in the masthead, the rail, and the page at once. The decision now: the rail belongs to lessons, where it shows position; Start has none. One lesson index on Start, grouped "The stack" (O1, O2) and "Provider code and engineering" (O3 to O7). The masthead carries no Explore until Explore is a distinct view (fn-3v8), and Listen and Field guides sit under one quiet Resources disclosure (historical: shipped as a9a5ece; since revision 12 the masthead carries two quiet icon links, Audio deep dives and Visual field guides, and no disclosure; the current shape is under "Start: one introduction and one index"). The book labels stay as quiet group labels. Shipping as fn-ga6.

Explore page (fn-3v8)

A chapter entry with `kind: 'map'`, `diagram: 'overview'`, a new group, and a `mode: 'explore'` flag that makes the renderer skip the lesson frame. The stack map with the developer and request paths, every component open, the drawer as its explanation surface. Under the map: the owner legend and one line of links to the lessons that teach each region. Routes `#explore`, `#explore/request`, `#explore?detail=flux`; the existing chapter-to-renderer and detail-coverage tests cover it without new test code. "Open in Explore →" from the stack lessons carries the current path and selection. The boundary, fork, and seam diagrams are shaped by their lesson's controls and can join later through a picker.

Start: one introduction and one index

Decided Two reviews in one day converged. The first-visit review (4680b0e) put the SPI definition, the two doors, the dated review line, and the introduction's entry on the first screen. The reading-flow review then measured the result: the Watch button scrolled the page 1,132 pixels to play the video, the doors and cards and figures were four indexes of one curriculum, and the opening introduced the running example's construction before the course's purpose.

The shape now (fn-ga6, then the same-day refinement with the owner, fn-9g2 and after): headline "Understanding the Azure SPI machinery", a title without a full stop, with a subheading naming the three parts; under them, side by side at desktop width so that together they span the same width as the cards below, the machinery strip on the left (three meshing gears in the owner colors with the loop arrow, cropped to the gears; prompt and original in docs/reference/start/) and three short paragraphs on the right, the strip stacking first on a phone. The paragraphs that say how the parts work together: the stack and what it runs; the Azure provider that shared code calls through the Service Provider Interface (SPI); the forks that maintain that provider while taking upstream OSDU changes, with the workflows that bring those in, build images, and test against a running stack; and the OSDU Community Implementation (CIMPL), linked, as the reference the lessons compare against. The subheading names the parts, the strip shows them, the paragraphs relate them; the site's naming convention is left to the three cards. Then Introduction with Watch · 1 min, captioned "How the machinery is engineered" (a modal; the card carries a gear-and-play badge), and Listen · 2 min (the brief in the dock); "SPI means three things here" in one order used everywhere on the page, the stack, the interface inside each service, the engineering system, with titles of one length ("The Azure stack", "The service interface", "The engineering system") and each repository link aligned at the bottom of its card; the one grouped lesson index with questions short enough for one line; Go deeper at the end as a quiet two-column list of six links separated by thin rules, one column on a phone (a kicker naming the repository or the OSDU community in the owner color, a linked title, one line on what is there; cards were dropped in revision 16 so the page does not end on a fourth grid of boxes), and the lessons' documentation accordion hidden on Start. The masthead carries two icon links, Audio deep dives and Visual field guides; on a phone the labels are hidden from sight but not from screen readers, so the header no longer pushes the page past the screen. No rail on Start or on either supplement page; the rail is lesson navigation only. "OSDU on Azure" does not appear: that name belongs to the earlier GitLab-hosted implementation.

Still rejected: leading the page with the video and the brief. The video's own source checks correct its narration twice and the brief spends most of its two minutes on why OSDU exists; a modal from a Watch control gives the video an intentional surface without making it the opener. Open: re-record the brief from the approved script so it opens on SPI.

Visual field guides: the set a learner reads

Visual field guides: the set a learner reads

Decided An adversarial pass over all nineteen items (two course maps, eight native guides, nine posters) judged each on what it teaches and how a lesson uses it. Three posters go: the generated "supplied" posters (blueprint, permanent fork, continuous forking) needed correction lists longer than their takeaways ("Portions", "Configblob", "CITOPS LOCK-FILE", a cascade drawn backwards) and everything true on them is already a native guide or the contribution chain. A poster that must be read with a list of what it gets wrong teaches distrust of the page, not the subject. Their links resolve to the guide that carries the subject; the originals stay in the reference folder. The other six posters and all eight native guides stay: each is shown beside at least one lesson (the backing environment now beside lesson 06, where the handshake borrows it). The profiles guide was the closest call (one sentence drawn three times); it stays because the profiles myth and lesson 02 both point at it and the greyed rows do carry the point visually.

The page itself follows the Start and Audio deep dives treatment: a headline and subhead beside a drafting-sheet strip in the gear style, then sections grouped by the lessons the sheets sit beside rather than by how they were made: Maps of the course, The stack (lessons 01 and 02), One service (03), The fork (04 and 05), The seam (06), guides and posters mixed within each, so a learner finds the sheet for the lesson they are on. Gone: the by-need index (a second navigation whose needs were invented; blueprint sat under "Prove a change"), the kicker, the premise, the intro paragraph, the scope note, the counts in headings, the method copy ("built in HTML", "Print this page"), the second owner legend, and the documentation accordion. Two defects fixed while there: the timeline drew the CLI's last step after the CLI-exit line (the lifecycle doc's 45 to 50 minutes with about 30 for AKS and 10 to 15 for the Flux extension now fit before the exit), and the ladder's sixth box wrapped one word per line. The lesson-side change from earlier stands: a guide beside a lesson shows as a title and one line that expands in place.

Revision 22 brings the page level with the rest. A section row is the Go deeper shelf grammar: the sheet badge, the title, the lesson chips, and a preview built from the sheet titles ("Where the familiar things live, Four owners, four boundaries, and five more") in place of a count. An open section is a contact sheet: a grid of cards, one per map, guide, or poster, each the same object (a kind label, the title, one sentence, and a thumbnail when the sheet is a poster). Pressing a card shows that one sheet under the grid; pressing another swaps it. Every sheet is still in the page, hidden, so a `?guide=` route, a lesson's Field guides row, and the lightbox with its poster checks work as before; the route opens the section and its card before scrolling. The stack section went from seven full sheets in sequence to seven cards and one sheet (fn-s30, PR #23, dd12a48).

Audio deep dives: two recordings, offered plainly

Decided The page listed four recordings with the twenty-two-minute orientation first, so a visitor who started at the top started on the framing and could stop before the two recordings the site most wants heard. Now the page offers the stack and the fork only, the

stack first and playing by default, as two cards with generated badge art (a gear wearing headphones, in the owner color), a kicker naming the lessons covered and the length, the title, and a one-line summary. The orientation and the brief stay in the data: Start's Listen cue and the lesson cues still play them, and their published routes still render. The page's framing copy (a kicker, a premise, an intro paragraph, a scope note, a documentation accordion, and an explanatory paragraph under Markers and under Transcript) is reduced to a headline, a subhead beside a headphones strip in the site's gear style, and one line in the player that names the recording's origin and says the marker's source check is where narration and documentation part ways. The page shares Start's 1000-pixel measure. Shipped as fn-z08, commit d9e84f6.

The polish review: corrections first, presentation later

Decided A sixth review of Start, Audio deep dives, and Visual field guides found the remaining tells concentrated in copy the site wrote for itself, not in the new opening: marker titles borrowed from the recordings ("Memory for an amnesiac runner"), count formulas standing in for summaries ("Three contracts, one credentialed job, five steps"), a subhead that promised markers open lessons when they seek the audio, and four claims that were plainly wrong or ahead of the code (a tag labelled as a digest on the round-trip map, "the forks own only their configuration", "State lives in Azure PaaS" one marker before the in-cluster middleware, every middleware password "mirrored to Key Vault" when only Redis and Elasticsearch are). All of that is corrected in place, in the site's own voice, without touching a transcript. Two behaviors changed with it: browser Back now returns to the marker or guide the reader left from instead of the top of the page, and on Audio deep dives the dock hides while the full player is on screen so there is never a second pause button. A poster's checks now travel into the lightbox with the image, and the two seam posters say in their kicker that they draw a target design and when their status was last reviewed.

The three follow-ups the review left are done, in the order it asked. Borrow, Prove, Restore and The Backing Environment are site-built posters now (sources in `docs/reference/posters/*-site.html`): every statement on them is as built as of September 2026, and what the design guides describe ahead of the code sits in a dashed "Not yet built" band on the poster itself, so no correction list is needed under either (fn-7gy). The field guides page is five lesson-group disclosures, the maps open and the rest closed, a `?guide=` route opening the group it lands in; a poster's caption is its title, one sentence, a scope note, and one lesson link, with takeaways, checks, and sources folded beneath; the enlarged poster fits the dialog until Zoom in is pressed; the page fell from about 12,600 to 2,400 pixels on desktop (fn-sys). Three replacement scripts are drafted for approval in `docs/`, each opening on the engineering question, chaptered to the site's markers, about 28 minutes for each deep dive and two for the video, with a table of what each says differently from the recording it replaces; the recordings themselves are a new bead (fn-y2o) and the transcripts stay until one is replaced (fn-45p). Two of its suggestions are declined for now: renaming the native guides away from their titles (the infographics and their names are the part of the page the owner most wants kept) and redrawing the sni

and their names are the part of the page the owner most wants kept), and redrawing the split up timeline as an untimed sequence (its bars now match the lifecycle document's minutes, and the legend already separates observed durations from the CLI exit).

Audio

45 of the 74 marker routes carry `?detail=`; each opens the drawer, as shipped. The clean version of "Listen to this lesson" is worth keeping for later: one three-minute narration per lesson recorded from an approved script, where the script is the lesson's claims. Then the audio cannot drift from the page, because it is the page. Decision Q6 below.

Section 7

Litmus questions and the pilot

The north-star test: after each lesson, can the learner answer its question without opening the drawer? Audited against today's content. The story layer is on the page everywhere; the two soft spots are named.

LESSON	AFTER IT, CAN YOU...	WHERE THE ANSWER LIVES	DRAWER-ONLY?
01	draw roughly where AKS sits in the complete stack?	Headline; the map's two halves under one stack boundary; claim 1.	No
02	explain how that stack gets created?	Six stages as the one control, each with owner, title, copy, timing, and its explanation in view; three statements.	No
03	point to where shared OSDU code ends and Azure provider code begins?	The two blocks with owner labels; the example note states the fallback; claims 1 and 2.	No Since PR #6 the cache-throw sentence is in claim 1's reason and the Cache down trace shows it on the map.
04	explain which code you own versus which you consume?	The ownership table; the what-if switch; claims 1 and 3.	No
05	describe how a change gets from source to the running environment?	Five moments; the hops; claim 3. The fix reaching main is told in 04's hops.	Split across 04 and 05 The bridge line fixes this.
06	say what proves a candidate image in the stack and what gives the slot back?	Six hops; the state panel; the example note; claims 1 and 2.	No

07	identify the common misconceptions?	The page, by theme, with a check each.	No
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The pilot (fn-29r)

Four or five engineers who know OSDU and not Azure SPI, on lessons O1 to O3 as published, watched, no scores. Three questions, asked on lesson O1 and repeated on whichever of O2 or O3 they reach:

1. Without clicking anything, what are the three things this lesson is trying to teach you?
2. After using the claims, can you explain what the map is showing without opening the inspector?
3. When you click "How we know", does the drawer feel like useful evidence, or like you have left the lesson?

Also watch: do they know when the lesson is finished, do they open the running example unprompted, do they find the receded components clickable, and do they notice the map keeps its state when they close the drawer. Notes go in a dated doc under docs/ ; the go or no-go lands on the roll-out epic. If they need the drawer to answer question 2, some story content is still in the wrong layer, and the roll-out waits until O1 is fixed.

Section 8

The plan in beads, and how work runs

The beads are the plan of record; bd show <id> has the acceptance criteria. This is the sequence and the intent behind each.

fn-5rl
epic · closed

Shipped. Lesson O1 reference implementation and the evidence drawer. PR #4, deployed 13 September.

fn-d5c
task · closed

Shipped. Round-four tunes: short claim headlines, quiet rail, the drawer content rule in AGENTS.md. On main, 4b7ed98.

fn-cca
task · closed

Shipped. Lesson O2 in the frozen grammar. Keelson beads-work run with the Copilot coding agent,

approved plan with two corrections, review loop, then an orchestrator fix pass (dropped an added playwright dependency and browser test, simplified the Look closer label, trimmed the notes). PR #5.

fn-bye

task · closed

Shipped. Lesson O3 with the cache-down trace. Same route; the fix pass laid the SPI map in two columns, restored the original carry-forward sentences, and fixed the hover label on dark nodes. PR #6.

fn-2bc

review · closed

Round five, lessons O1 to O3 as implemented. Seven follow-ups below.

fn-kbc · fn-loo

bugs · closed

Shipped. The two factual corrections: lesson O3's Redis inside AKS versus Table Storage outside, and lesson O2's readiness signals without a false order. Beads-work runs.

fn-mjx · fn-5pw · fn-wes

bugs · closed

Shipped. Drawer visible from its opener at 800 px tall, focus into the drawer on first open, and a saved evidence URL restoring a claim rather than a trace. One run, since they share `main.js`.

fn-p2j · fn-e7u

tasks · closed

Shipped. Evidence that proves the current claim (the POM and Dockerfile for "one image"; an ownership field), and the learner-facing copy edit. One run.

fn-29r

task · alongside

Pilot lessons O1 to O3 with four or five engineers. Runs alongside the roll-out; its findings land as beads. Owner: you; the protocol is section 7.

fn-fm9

epic · closed, not doing

Navigation frame. Closed after the fourth round: the current frame works once the rail is quiet. Reopens only on pilot evidence.

fn-3v8

epic · closed, not doing

Explore page. Closed 14 September: revision 17 settled that Explore mode does not earn its place on

Explore page: closed 14 September: revision 11 settled that explore must also not carry its place on the map, and a separate page is the same idea in another room.

fn-8kx

epic · closed

Roll the lesson shape across O2 to O6. Lessons O2 and O3 carry the claim grammar; O4 to O7 carry the shared frame; lesson O6 is driven by its run steps (fn-dsx, PR #24). Decision 14 September: lessons O4 and O5 keep their example hops as the workspace control and are not migrated to claim selectors unless the pilot shows they do not land.

fn-nkh

epic · closed

The hands-on track of section 5: the band rule, the `tryIt` field and renderer, bands on O1 to O3 (O2's Azure route a marked preview). Decision 14 September: no further preview bands; the O4 and O5 bands are the outcome of the walkthrough and sit in fn-dt3's acceptance.

fn-nh5

decision · closed

The interaction-hierarchy rules, section 1, recorded as policy in AGENTS.md; closed in revision 15.

How the work runs

- **One implementer per bead, briefed from the bead** (goal, scope, context pointers, acceptance, verify commands, forbidden list, report shape). A bead runs as a Keelson beads-work run in its own worktree: plan, approval gate with the orchestrator's corrections folded in, implementation, an adversarial review loop with fixes, then a draft PR.
- **The orchestrator reviews the PR in the browser** (check output, screenshots at 1440 and 390, DOM probes for the invariants), makes a fix pass on the branch, marks it ready, and merges it. The local check is the gate; the repository runs no CI on pull requests.
- **Small orchestrator edits go to main directly.** Every push to main keeps `npm run check` green and renames no route, because Pages deploys on push. Two or three runs go in parallel when they touch different files; a shared file means a merge with conflicts the orchestrator resolves.
- **Beads close at push** with the commit as the reason; the tracker state is committed alongside.

Section 9

Decisions

Settled ones carry their outcome. Open ones carry a recommendation.

Q1 Ship lesson O1 alone before redesigning anything else?

Settled: yes, and the interaction model is no longer under review. Shipped in PR #4; the reviewer's fourth round confirmed the concept. Energy goes to expressing O2 and O3 in the same grammar.

Q2 Do lessons O2 and O3 wait for the pilot?

Settled: no, and both shipped. The sequencing of O4 to O6 is decided below, under "Do lessons O4 to O6 still wait for the pilot?".

Q3 Does the navigation change?

Revised in revisions 11 and 12: yes, once. The rail belongs to lessons and Start has none; the masthead carries two quiet icon links (Audio deep dives, Visual field guides) and no Explore until Explore is a distinct view; book labels stay as quiet labels. The earlier "Settled: no" (fn-fm9 closed after round four) was reopened by the reading-flow review, section 6.

Q4 Must every lesson have three claims?

Settled: no. Two to four, set by the content; the tests enforce the range. What stays constant is claim, visual evidence, inspection, carry-forward.

Q5 Does a deep link with `?detail=` open the drawer?

Settled: yes. It opens without taking focus from the headline.

Q6 Do receded components stay clickable?

Settled: yes. The affordance shows on hover and keyboard focus.

Q7 Does Explore map earn its place?

Settled in revision 17: no. It changed weight and affordances only; the owner read it as a hidden layer. Removed.

Q8 Should stepping through lesson O6 open the drawer?

Settled in revision 23: no. The six run steps are the workspace control (fn-dsx, PR #24). Selecting one moves the lock and pod, focuses its component, and shows the hop line in place; only its "How we know →" opens the drawer, and closing the drawer keeps the step.

Q9 What does closing the drawer do to the URL?

Settled: it removes `detail`. One Back press after closing does nothing visible; the alternative would make reload and the URL disagree with the screen.

Q10 Does Explore start with the stack map only?

Recommend yes. The other three diagrams are shaped by their lesson's controls; a picker can follow.

Q11 Do we commission per-lesson narration from the claims?

Recommend after the pilot. If the claims survive unchanged, they are the script, and "Listen to this lesson" becomes truthful and can move up.

Q12 Do the candidate headlines in section 4 replace the current ones?

Recommend lesson by lesson during the roll-out, keeping any current headline that already is the sentence to remember (O6 and O7). Claim headlines are separate and every claim gets one.

Q13 Does lesson O7 stay a lesson?

Settled in revision 16: it stays, presented as a field check. The round trip ends on O6 ("That is the round trip"); O7 keeps its slot, route, and outcomes, and its 22 assumptions render as compact rows rather than a wall of cards.

Q14 Does the site get a hands-on track?

Settled: yes, as an optional Try it band per lesson, after the bridge, collapsed by default, no tracking. The product intent's ban on required exercises and live execution stands; the band describes and the workstation runs.

Q15 Does the track follow the lesson order?

Settled: the lesson order stays; each activity states its prerequisites and can be started once they are met. A reader without Azure enters through browser inspection, workstation preparation, and the public GitHub route; O2 and O6 need Azure resources billed separately; O6 needs the fork from O4 and the environment from O2.

Q16 Personal account or organization?

Settled: the personal account is the written path. The repositories need no organization; the docs' "organization" wording is rewritten in the band and the org variant is a note.

Q17 Do lessons O4 to O6 still wait for the pilot?

Revised: no. The round-five fixes and the Try it band come first, then O4 to O6 with bands. The pilot runs alongside and files beads.

Q18 Which environment names appear in a band?

Settled: only the example partition and placeholders. `opendes` as the illustration, `<name>` for the environment; the site names no environment because each learner creates their own with `--name`. None of ours.

Q19 Does a recipe ship before someone has run it?

Settled: no, with one stated exception. A browser-only route ships once it is verified in the browser. An Azure route ships only after its walkthrough; until then it may appear only as a clearly marked preview, which is what lesson O2's "not walked" label is today. Each band records the tested CLI release, stack ref, template commit, shell, and date. Lesson O6's band stays planned until its descriptor and workflow route is demonstrated end to end.

Q20 Does the site name a reference implementation?

Settled: yes, CIMPL, introduced on the start page and used selectively. No new lesson, no global CIMPL/Azure toggle, no substitution table on a lesson surface. Prototype first (fn-jy3): the introduction, one comparison sentence in lessons O1 and O3, and the optional paired lookup on O3. Lesson O4's opening, the familiar-things guide, lessons O5 and O6, and a field guide follow if the prototype reads well.

Q21 Can a learner merge their own provider change alone?

Settled: not under the shipped rules. The band makes observing the change's validation a complete stopping point and names a second reviewer as the prerequisite for merging. The rules are not loosened to make the example work.

Section 10

Changelog

- **Revision 23, 14 September.** Lesson O6's run steps become its workspace control (fn-dsx, PR #24, 62be86e, built by a subagent from the bead and verified in the browser at desktop and 390 pixels). Owner decisions recorded: lessons O4 and O5 keep their hops and are not migrated to claim selectors (fn-8kx closed); no further preview bands, the O4 and O5 bands are the walkthrough's outcome (fn-nkh closed into fn-dt3); the Explore page is not doing (fn-3v8 closed). Filed fn-kwp, a reading-flow check on lessons O3 to O6, blocked on this change. The tracker now holds fn-kwp and the owner's three tasks.
- **Revision 22, 14 September.** Owner: bring Visual field guides level with the other pages (fn-s30, PR #23, dd12a48, built by a subagent from the bead and verified in the browser at 1000 and 390 pixels) and drop the notebook and download links from the deep-dive player (fn-dhu, PR #22, b5de2c0). Groomed: fn-mtw closed after a desktop and 700-pixel check of lesson O3's lanes and six hops in both variants; fn-wqb closed, its review

having shipped in revision /. Open for the owner's decision: whether the claim-selector migration for lessons 04 to 06 (fn-8kx, fn-dsx) is part of done, whether the 04 and 05 bands ship as marked previews ahead of the walkthrough (fn-nkh, fn-dt3), and whether the Explore page (fn-3v8) closes as not doing now that Explore mode is gone.

- **Revision 21, 14 September.** Owner: carry the footer through every learning lesson (PR #21, a089057, done directly). The Go deeper shelf, the scope caption under the figure, the Sources row, and the light What you can now say now depend on a lesson being a learn chapter, not on the claim grammar, so lessons 04 to 06 lose the standalone example row, the full guide figures and poster thumbnails under the map, the dark outcomes panel with its inline Next paragraph, and the Go deeper in the documentation disclosure; lesson 07 gets the Sources row. Lesson 06's round-trip choices render as light cards (their text had been wrapping one word per line under the outcomes grid). The claim-selector migration for 04 to 06 (fn-8kx, fn-dsx) is unchanged.
- **Revision 20, 14 September.** Owner notes, done directly (PR #20, 2cbcd07). The dark next-lesson bar is replaced by a footer pager on every chapter: position, Previous, and a Next lesson card carrying the kicker, title, and question; lesson 07 keeps Back to the start and the supplements a plain Next. Inside Try it the Tested with table becomes one muted line naming the walked-on date and stack ref; the full record stays in content as publication evidence (rule 31 unchanged in substance). Rule 11 and the state line are revised.
- **Revision 19, 14 September.** The reviewer's follow-up on the shelf, done directly (PR #19, 41d29ed): "calmer, but a little too much like a reference table". Try it becomes a banner with a drawing for the activity (workstation; workstation and cloud; code window and magnifier), the title, a one-sentence outcome, and an Optional qualifier. Each Go deeper row gets a badge in the style of the generated artwork (path through three owner nodes, headphones with gear cups, folded sheet with magnifier, open notebook), drawn as inline SVG in badges.js because the originals were generated from prompts. Previews are counts from the content instead of title lists, titles appear on opening, and the play chips move inside the audio row. Taken deliberately for the footer only, against the earlier Learning review's no-icons advice for the lesson body. Rules 1 and 14, the Try it summary, and the shipped-shape list are revised.
- **Revision 18, 14 September.** The lesson-bottom review, one beads-work run (PR #18, 412c5f6). Lessons 01 to 03 end with the Try it row and then one Go deeper shelf of closed disclosure rows: Follow the example, Hear it explained (play chips visible beside the row), Field guides (Expand guide; a poster as title, sentence, and View poster), Sources. Previews come from the content. The scope sentence becomes a caption under the figure. Deep links open the rows they land in, and Back reopens the rows the reader left open. On lesson 01 at 1280px the collapsed band goes from 891px to 261px and the page from 3,551px to 2,658px. Rules 1, 5, 11, and 14 and the shipped-shape list are revised; lessons 04 to 06 are unchanged until they roll into the grammar.
- **Revision 17, 14 September.** Two owner notes and one reviewer note, done directly (PR #17, f5669bf). The Explore map toggle is removed with its policy state; focus, receding,

and the hover or focus affordance remain, plus a hover-none rule for touch. Closing the drawer keeps the selected component and the map state it drives, which fixes the lesson 06 reset the reviewer hit; a claim, stage, or lesson change still clears it. The open-summary marker rotation is scoped to a marker class, so the lesson chips in a Visual field guides section stop spinning when it expands. Rules 6, 9, and 13 and the pilot watch list are revised; the lesson 06 step-selector proposal is filed as fn-dsx under the 04 to 06 epic.

- **Revision 16, 14 September.** The Learning design review, taken in two beads-work runs. PR #15 (d1df9b9): the idea selectors move inside the map workspace under the title row with the selected explanation directly beneath them and the map under that, the "1 / 3" counter is removed, lesson 02's statements follow its stage-driven workspace, and the inspector loses its Inspecting, Context, Owned by, What it is, and Why it matters here labels in favour of the name, a context line with an optional owner, two plain paragraphs, More, Verify it, and Sources. PR #16 (9ae2cab): Go deeper becomes a two-column ruled list, and lesson 07 becomes "Field check: the assumptions that cause trouble" with each myth as a row (assumption, one-sentence correction written from its existing text, and a closed Check and sources disclosure); the collapsed page at 390x844 falls from 10,262px to 5,268px. Not taken from the review, by its own advice: no progress, no checkmarks, no quiz, no icons, no gradients, no further copy rewrite. Rules 1, 2, and 9, the Start paragraph in section 6, the lesson 07 entry in section 4, and the lesson 07 decision are revised to match.
- **Revision 15, 14 September.** The polish review's three follow-ups shipped: the two seam posters redrawn as built with a dated Not yet built band (70030f9), the field guides page collapsed into lesson-group disclosures with lean poster captions, a fitting lightbox with Zoom in, profile-state text, and a fuller timeline description (4317025), and three replacement scripts drafted for approval (e2679b9), then revised against the owner's script review (8054bb8): the two token lifetimes kept apart, the credential split by job (App tokens, GITHUB_TOKEN with package write, GitHub OIDC), the deploy lane's adoption status stated where the lane is introduced and dated, the CLI and Flux overlap, profile scope without cost claims, bootstrap inputs in the namespaces that consume them, no guarantees in the closings, the video opening on the three parts, and narration-only files in docs/narration/ as the only input a generator receives. Recording (fn-y2o) now also covers the approved brief that Start's Listen cue still plays in its old form. Backlog groomed: fn-nh5 closed as recorded policy, fn-3v8 to backlog, fn-nkh behind fn-8kx, fn-y2o filed for the recordings.
- **Revision 14, 13 September.** Reconciled after the verification report: the state line names 31947da, the navigation decision and the Resources disclosure are marked historical with the current masthead named, the drawer template is dated, the round-five and evidence beads read closed, and the recipe policy states its one exception. The storage poster image now carries its new title and wording, the namespace poster its new title, the enlarged poster shows its origin and status above the image, and correction bullets no longer split across columns. The polish review's corrections applied: marker summaries

and titles rewritten plainly, the duplicate Source check prefix removed, the round-trip map pins a digest, the owners guide names what controllers and operators own, the familiar guide separates the schema loader from the schema data and scopes the Key Vault copies to Redis and Elasticsearch, the labels guide drops its count, the two seam posters declare themselves target designs with a review month, poster checks travel into the lightbox, the listen subhead says markers seek and links navigate, the Start lesson questions and Go deeper notes are shortened, the concept cards read as sentences. Back restores the reading position; the dock hides behind the full player. Commit 31947da.

- **Revision 13, 13 September.** Start rebalanced after a second look at the live page: the strip moves under the subhead and is cropped to the gears, the paragraph is shortened to what the subhead does not already say, the three meanings take one order everywhere (stack, interface, engineering) with same-length titles and bottom-aligned repository links, the Watch card is captioned "How the machinery is engineered", the leftover documentation accordion is hidden on Start, long lesson questions are trimmed, and the masthead fits a 390-pixel screen. Commit 032d2e0. Then the opening became three short paragraphs written with the owner, one idea each, with the naming convention left to the cards, set beside the strip so the opening spans the width of the cards, under the title "Understanding the Azure SPI machinery" (no full stop), and Go deeper redrawn as six source cards instead of a row of links. Audio deep dives then reworked to offer the two deep dives only, with generated artwork and the framing copy cut to a subhead and one player line. Visual field guides then reduced to what teaches: three generated posters retired, the by-need index and method copy dropped, a drafting-sheet strip added, the timeline and ladder defects fixed. Commits 8eb90fb and after.
- **Revision 12, 13 September.** Start refined with the owner after the round shipped: machinery headline and subhead, the intro rewritten around the three parts with CIMPL linked in place, a generated machinery strip and a gear-and-play Watch badge, the three meanings reordered with repository links, sources reduced to a plain Go deeper section, masthead icon links (Audio deep dives, Visual field guides), and no rail on the supplement pages. Commits 4954d15 to 4d78d62.
- **Revision 11, 13 September.** The reading-flow review of Start and lessons 01 to 02 (docs/start-lessons-01-02-reading-flow-review-2026-09-13.md). Navigation reopened: no rail on Start, one grouped index, no masthead Explore, a Resources disclosure. Grammar loosened: question and goal not displayed, the example as optional depth after the exit, a lifecycle lesson may use its stages as the main control, selection is idempotent. The interaction rule replaces "nothing moves the page". Status blocks refreshed to what is live; the lede no longer accumulates revision history. Beads fn-ga6, fn-bhr, fn-qwy, fn-wo6, all shipped the same day (main a442848).
- **Revision 10, 13 September.** Start page first screen rebuilt after a first-visit review (fn-mdv): SPI defined in the intro, dated review line, two doors and the introduction's entry in the Start panel, the lower doors section folded in, "Optional" dropped. Leading with the

media and hiding the rail were considered and rejected; the reasons are in section 6. The environment name `dev1` is gone from the site (fn-ix3): commands use the marked placeholder `<name>`, prose says your environment or the stack, `opendes` stays as the running example partition, and the published `dev1-slot` route aliases to `stack-slot`. Two raw-HTML fields that swallowed `sha-<commit>` are escaped (fn-57h). Beads JSONL exports are no longer tracked in git.

- **Revision 9, 13 September.** PRs #10 to #13 merged: claim-specific evidence with an owner row only when authored, the learner-facing copy edit, the CIMPL prototype, and the Try it band renderer. Recipes for lessons O1 to O3 are live; lesson O2's Azure route is marked not walked. Status and plan updated; no design change.
- **Revision 8, 13 September.** CIMPL named as the reference implementation after the community-baseline review (fn-jy3): a reference paragraph in section 4 and a decision. PRs #8 and #9 merged (readiness signals; drawer fit, first-open focus, claim and hop route qualifiers).
- **Revision 7, 13 September.** Try it revised after the source-checked plan review (fn-wqb): prerequisites instead of any-order, activity names on the collapsed line, access labels instead of free or billable, effort and waiting separated, the dry run's Azure requirement, the lesson O3 image observation, both sync outcomes, the review-rule and package-visibility findings, lesson O6 kept planned until demonstrated, tested versions recorded per recipe. One current policy per question; Practice added to the layers table; section 8 describes the run-and-merge workflow as it is. Lesson O3's boundary fix merged (PR #7).
- **Revision 6, 13 September.** Try it track proposed; fifth review of lessons O1 to O3 recorded with seven follow-ups; O4 to O6 unblocked from the pilot.
- **Revision 5, 13 September.** Lessons O2 and O3 shipped as Keelson beads-work runs (PRs #5 and #6). Compared with the lesson O1 pass through Codex, the plans were stronger and the diffs needed fewer fixes; the review loop's one bad call was adding a dependency the project forbids, caught at the fix pass. The lesson O1 review had found ten defects the implementer's own checks did not.
- **Revision 4, 13 September.** Round-four tunes: short claim headlines, quiet rail, drawer content rule in AGENTS.md; navigation frame closed.
- **Revisions 1 to 3, 12 to 13 September.** Lesson O1 reference implementation and the evidence drawer (PR #4); the interaction hierarchy settled.

